



VECTOR SPACES

Deepthi Rao

Department of
Science & Humanities

CLASS 2 : CONTENT



- Echelon form of a matrix
- Row reduced Echelon form of the matrix
- Pivot variables and Free variables
- Special solution

VECTOR SPACES :

ECHELON FORM U AND ROW REDUCED FORM



A rectangular matrix is said to be in echelon form if it has following characterizations

- i. All the zero rows are below the non zero rows.
- ii. Each pivot (non-zero leading entry) lies to the right to the pivot in the row above(This produces the stair-case pattern).
- iii. All the entries in a column below the pivot entry are zero.

The matrix is said to be in row reduced form R , if in addition to the above, the matrix has following additional characterization

- iv. Pivot (it should be 1) is only non-zero entry in its column

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PIVOT VARIABLES AND FREE VARIABLES



Consider $Rx = 0$

$$\text{i.e., } \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The unknowns are divided into two groups

- i. Pivot variables: which corresponds to **columns with pivots**.
- ii. Free variables: which corresponds to **columns without pivots**.

From the above example

First and third columns contain the pivots, so **u & w** are the **pivot variables**.

Second and fourth columns do not contain pivots, so **v & y** are **free variables**.

RANK OF A MATRIX

Rank of a Matrix Definition: The rank of a matrix A is the number of nonzero rows in the echelon form U of A and is denoted by $\rho(A)$ or simply r .

Note : If A is a matrix of order $m \times n$ then its rank $r \leq \min(m, n)$.

$$\text{Ex: 1} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -1 \end{bmatrix}; \rho(A) = 2$$

$$\text{Ex: 2} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}; \rho(A) = 1$$

VECTOR SPACES

ECHELON FORM U AND ROW REDUCED FORM

Eg:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} = U, \quad \text{Echelon form}$$

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 6 \\ 0 & 0 & 0 \end{bmatrix} = U, \quad \text{Echelon form}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} = R, \quad \text{Row reduced Echelon form}$$

Marked elements are the pivots.

To find the most general solution to $Rx = 0$ (or equivalently to $Ax = 0$) we may assign arbitrary values to free variables.

The pivot variables are completely determined in terms of free variables

$$v \text{ and } y \Rightarrow Rx = 0 \Rightarrow u + 2v - y = 0 \Rightarrow u = -2v + y$$

$$w + y = 0 \Rightarrow w = -y$$

Special solutions

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Most general solution or complete solution

The best way to find all solutions to is from the special solutions

$$x = \begin{bmatrix} u \\ v \\ w \\ y \end{bmatrix} = v \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

The complete solution is the linear combination of 2 special solutions.

VECTOR SPACES

ECHELON FORM U AND ROW REDUCED FORM



E.g. : For every c , find R and special solutions to $Ax = 0$, where

$$A = \begin{bmatrix} 1-c & 2 \\ 0 & 2-c \end{bmatrix}$$

Solution: If $c = 1$ then $A = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}$

$$R_2 \rightarrow R_2 - 2R_1$$

$$A \sim \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} = U$$

$$R_1 \rightarrow \frac{1}{2} R_1$$

$$R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$R_x = 0 \Rightarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow y = 0 \text{ \& } x \text{ is free variable}$$

VECTOR SPACES

ECHELON FORM U AND ROW REDUCED FORM

$$x = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

If $c = 2$ then

$$A \sim \begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix} = U; R_1 \rightarrow -R_1$$

$$\sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

y is the free variable

VECTOR SPACES

ECHELON FORM U AND ROW REDUCED FORM



Now $x - 2y = 0 \Rightarrow x = 2y$

Special solution : $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

If $c \neq 1, 2$ then no special solution

For E.g.: If $c = 0$ then

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} = U; R_1 \rightarrow R_1 - R_2$$

$$A \sim \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = R$$

$$Rx = 0 \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x = 0, y = 0 \quad (\text{No special solution})$$



THANK YOU

Deepthi Rao

Department of Science & Humanities